

KETAPANG RAHADI USMAN, Indonesia

WMO: 966150

Lat: 1.8174S

Lon: 109.9620E

Elev: 46

StdP: 14.67

Time Zone: 7.00 (E07)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	73.5	74.2	69.5	108.5	82.6	70.4	112.1	81.7	16.5	83.0	13.6	81.9	3.9	110	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.6	92.1	79.2	91.3	79.2	90.6	79.1	81.5	88.6	81.0	88.0	80.6	87.7	7.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.6	153.6	85.0	79.1	151.1	84.6	78.7	149.0	84.3	45.2	88.9	44.7	88.5	44.2	88.2	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	12.5	11.1	69.1	95.6	3.9	3.0	66.3	97.7	64.0	99.5	61.8	101.1	59.0	103.3
			66.5	83.6	4.1	2.8	63.6	85.6	61.2	87.2	58.9	88.8	55.9	90.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	82.2	82.1	82.7	83.0	83.5	83.2	82.2	82.3	82.4	82.2	81.9	81.7	(6)
(7)		DBStd	1.66	1.76	1.52	1.53	1.44	1.52	1.66	1.79	1.53	1.59	1.58	1.42	1.61	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11841	998	899	1013	989	1037	996	999	1001	971	998	957	983	(10)
(11)		CDD65	6366	533	479	548	539	572	546	534	536	521	533	507	518	(11)
(12)		CDH74	67141	5678	5075	6032	5959	6346	5961	5468	5470	5368	5501	5097	5186	(12)
(13)		CDH80	23871	1783	1636	2160	2223	2469	2329	2054	2094	2000	1958	1629	1536	(13)
(14)	Wind	WSAvg	4.4	4.3	4.0	3.2	3.1	4.0	4.9	5.9	6.6	6.0	4.0	3.3	3.6	(14)
(15)	Precipitation	PrecAvg	128.1	13.7	9.8	10.4	12.2	9.4	7.8	5.9	3.8	6.3	11.9	17.6	18.6	(15)
(16)		PrecMax	196.0	25.6	21.0	26.9	23.9	22.1	14.6	15.8	14.1	17.9	26.3	30.0	32.9	(16)
(17)		PrecMin	86.3	4.1	3.0	1.6	5.3	1.3	2.0	0.2	0.0	0.4	1.2	7.4	9.8	(17)
(18)		PrecStd	24.6	6.6	4.6	5.4	4.7	5.0	3.9	4.8	4.2	4.6	6.5	6.3	6.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.6	90.4	91.2	92.4	93.3	92.8	92.4	92.3	92.1	91.8	90.9	90.0	(19)
(20)			MCWB	79.5	79.4	79.2	80.4	79.8	79.4	78.9	77.8	77.9	78.6	79.4	79.6	(20)
(21)		2%	DB	89.0	89.2	90.0	90.8	91.7	91.5	90.7	90.9	90.6	90.3	89.6	88.9	(21)
(22)			MCWB	79.3	78.9	79.4	80.1	80.0	79.3	78.6	77.8	78.0	78.9	79.3	79.4	(22)
(23)		5%	DB	88.0	88.2	89.1	89.7	90.6	90.4	89.6	89.6	89.5	89.2	88.5	87.8	(23)
(24)			MCWB	79.1	78.6	79.3	79.8	80.0	79.3	78.3	77.7	78.0	78.8	79.1	79.1	(24)
(25)		10%	DB	86.9	87.2	88.1	88.7	89.6	89.3	88.3	88.4	88.3	88.0	87.3	86.6	(25)
(26)			MCWB	78.8	78.4	78.9	79.7	79.9	79.2	78.0	77.6	77.9	78.6	78.9	78.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.6	81.0	81.4	81.8	81.9	81.7	81.1	80.6	80.4	80.7	81.0	81.0	(27)
(28)			MCD B	87.7	87.4	88.3	89.3	89.3	89.0	88.6	88.0	88.1	88.4	87.6	87.3	(28)
(29)		2%	WB	80.5	80.1	80.4	81.0	81.4	80.8	80.1	79.6	79.8	80.0	80.3	80.3	(29)
(30)			MCD B	86.7	86.4	87.3	88.3	88.8	88.4	87.4	87.2	87.5	87.5	86.8	86.2	(30)
(31)		5%	WB	79.9	79.6	79.9	80.5	80.8	80.1	79.5	79.0	79.2	79.6	79.8	79.8	(31)
(32)			MCD B	85.8	85.8	86.6	87.7	88.2	87.6	86.9	86.7	86.8	86.8	86.1	85.5	(32)
(33)		10%	WB	79.2	79.1	79.4	80.0	80.3	79.7	78.8	78.3	78.7	79.1	79.2	79.3	(33)
(34)			MCD B	85.0	85.1	85.9	86.7	87.4	87.0	86.4	85.9	86.1	86.0	85.3	84.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	10.4	11.2	11.7	12.6	12.9	13.1	14.0	13.3	12.3	11.1	9.9	(35)
(36)			MCDBR	11.2	11.7	12.3	13.0	13.9	14.3	14.6	15.1	14.7	13.9	13.0	11.5	(36)
(37)			MCWBR	4.0	4.2	4.3	4.6	4.7	4.8	4.8	5.1	4.9	4.8	4.6	4.3	(37)
(38)		5% WB	MCD BR	10.4	11.0	11.6	12.3	12.9	13.3	13.5	14.1	13.5	12.8	11.9	10.5	(38)
(39)			MCWBR	4.2	4.3	4.3	4.7	5.0	5.0	4.9	5.3	5.0	4.9	4.7	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.422	0.444	0.435	0.413	0.415	0.416	0.436	0.518	0.651	0.664	0.504	0.418	(40)	
(41)		taud	2.464	2.382	2.408	2.482	2.481	2.486	2.419	2.145	1.778	1.755	2.197	2.477	(41)	
(42)		Ebn at Noon	289	283	283	282	274	269	266	250	225	224	263	289	(42)	
(43)		Edh at Noon	37	41	39	36	34	33	36	49	73	75	48	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1555	1654	1671	1627	1635	1610	1647	1737	1663	1544	1518	1421	(44)	
(45)		RadStd	89	98	91	55	76	80	89	79	144	158	80	86	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page